

FitBuddy

Phase 6: Project Testing Phase

Smart Fitness and Wellness Application

1. Objective

To verify that FitBuddy works correctly and is usable, accurate, and reliable before release.

2. Types of Testing

Type	Focus
Unit Testing	Individual functions such as BMI calculation and recommendation logic
Integration Testing	Frontend, Python backend, and MySQL working together
Functional Testing	Each feature meets its requirement
Usability Testing	Ease of navigation and clarity for real users
Accuracy Testing	Recommendations match expected outputs for sample profiles
Security Testing	Login protection, input validation, SQL injection checks
Reliability Testing	Stable behavior under repeated use and multiple users

3. Sample Test Cases

ID	Test Case	Input	Expected Result	Status
TC1	User registration	Valid details	Account created	Pass
TC2	Login with wrong password	Invalid password	Access denied, error shown	Pass
TC3	BMI calculation	70 kg, 1.75 m	BMI approx. 22.9	Pass
TC4	Weight-loss recommendation	Goal: weight loss, moderate activity	Cardio workouts suggested	Pass
TC5	Log activity	Running, 30 min	Entry saved in database	Pass
TC6	View progress	User with 5 logs	Chart displays all entries	Pass
TC7	Empty form submit	Blank fields	Validation message shown	Pass
TC8	SQL injection attempt	' OR 1=1 --	Input rejected/sanitized	Pass

Note: the Status column is a template. Update it with your actual test results.

4. Defect Tracking

Bug ID	Description	Severity	Resolution
B1	Progress chart not refreshing after new log	Medium	Reload data after save
B2	Layout breaks on small screens	Low	Adjust CSS media queries

Sample entries; replace with the defects found in your own testing.

5. Conclusion

After fixing identified defects and re-testing, the application should satisfy functional, usability, accuracy, and reliability criteria.